

MissVARPath: A Meta-Classifer for Missense Variant Pathogenicity

Course CMP682 – Final Report

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Abstract

We build and evaluate a missense-variant pathogenicity classifier on a 21,872-row ClinVar/OpenCRAVAT dataset, comparing ten model families across five evaluation regimes. Our headline canonical result is **LightGBM at 0.801 macro-F1** on the 4-class (Benign / Likely-benign / Likely-pathogenic / Pathogenic) task and **CatBoost at 0.9895 macro-F1 / 0.9991 ROC-AUC** on the binary task. We additionally show that the classifier is not a pass-through of any single dominant in-silico predictor: removing DITTO (the highest-SHAP-importance feature) costs only 0.005 macro-F1, and a complete ablation of *all* learned predictor scores (AlphaMissense, REVEL, CADD, DITTO, MetaRNN, BayesDel, etc.) leaves the booster family at **0.730 macro-F1** on the 4-class task and **0.939 macro-F1** on the binary task, using only population allele frequencies, genomic position, and direct sequence-composition features. The methodology covers gene-stratified cross-validation (mitigating data circularity from overlapping ClinVar training data), a no-VEP / VUS-style scenario, k-mer + BLAST augmentation, two SHAP analyses, and a focused grid-search hyperparameter pass.

1 Introduction

The standard clinical-genomics pipeline for missense variants of uncertain significance (VUS) routes variants through a battery of in-silico predictors (PolyPhen-2, SIFT, REVEL, CADD, AlphaMissense, etc.) and then combines their outputs with population frequency data and ACMG/AMP rules. Each predictor disagrees with the others, the predictors have correlated training data, and most are themselves trained with at least partial overlap on ClinVar – the same database used to label our targets.

Three questions follow naturally:

1. Can a meta-classifier do better than any single predictor by combining their outputs with population and sequence features?
2. How much of any apparent improvement is data-circularity (predictors trained on ClinVar being evaluated on ClinVar)?
3. Can we predict pathogenicity from raw data alone – population frequency, genomic position, sequence composition – without relying on *any* predictor’s score?

This report answers each in turn. We train ten model families on a 21,872-row ClinVar/OpenCRAVAT dataset, evaluate under five regimes including a strict gene-stratified cross-validation, run two ablation experiments (single-predictor removal and complete predictor-score removal), and use SHAP to inspect which features the best models are actually using.

2 Data

2.1 Source

The dataset (`data/missense_dataset.csv`, 21,872 rows $\times$ 777 columns) is a curated set of missense variants drawn from ClinVar, annotated through OpenCRAVAT with 158 distinct annotator groups. The 4-class label (`clinvar_sig`) is perfectly balanced at 5,468 variants per class: Benign, Likely-benign, Likely-pathogenic, Pathogenic. The 2-class collapse pairs the “Likely” tiers with their certain counterparts, again perfectly balanced at 10,936 per class.

2.2 Preprocessing

Preprocessing (`src/preprocessing.py`) performs:

1. Filter to valid 4-class labels.
2. Drop label-leakage columns (39 columns, all under `clinvar_*` or `clinvar_acmg_*`, except the label).
3. Drop identifier and free-text columns (88 cols: rsIDs, HGVS strings, transcript IDs, disease names, PubMed IDs, URLs, etc.).
4. Coerce object-typed numeric columns; drop the rest as free text.
5. Drop high-missingness columns ($>50\%$ NaN, 207 cols).
6. Median-impute numeric, mode-impute categorical residuals; drop zero-variance.
7. Tidy column names; emit `target_4` and `target_2`.

Final base feature set: **21,872 rows $\times$ 208 features** with no missing values. The augmented variant adds 196 k-mer and 10 BLAST-derived columns (`src/sequence_features.py`, `src/blast_features.py`), yielding **21,797 rows $\times$ 414 features** for the subset of variants whose protein sequence could be fetched.

3 Methods

3.1 Train/test protocol

For every experiment we fix `RANDOM_STATE = 42` and use an 80/20 stratified holdout. The 80% training portion is then evaluated by 5-fold `StratifiedKfold` cross-validation. For gene-stratified runs we substitute `GroupShuffleSplit` (test holdout) and `StratifiedGroupKfold` (CV) on `gene_symbol`, ensuring no gene appears in both train and held-out folds.

3.2 Model suite

Ten models, three families:

- **Linear / shallow:** LogisticRegression (lbfgs, multinomial), ShallowNN-MLP (hidden=(128,), batch=128, max_iter=400, early stopping).
- **Tree ensembles:** RandomForest (400 trees), ExtraTrees (400 trees), XGBoost (600 / depth-6 / LR=0.05), LightGBM (600 / leaves=63 / LR=0.05), CatBoost (600 / depth-6 / LR=0.05).
- **Deep / sequence:** CNN1D, LSTM, RNN – each tuple of (Conv/Recurrent backbone $\rightarrow$ MLP head), implemented in PyTorch on the same standard-scaled feature vector.

3.3 Evaluation regimes

1. **Canonical kfold** on the base 208-feature parquet.
2. **Augmented kfold** on the 414-feature parquet (k-mer + BLAST added).
3. **Gene-stratified kfold** on the base parquet – enforces zero gene overlap between train and test.
4. **No-VEP / VUS scenario** – drop all 147 in-silico predictor-score columns, keep population frequencies + position + sequence features.
5. **Raw-only** (this report’s terminal experiment) – drop everything generated by an external model, tool, or algorithm, including conservation scores (GERP, PhastCons, PhyloP, SIPHY) and label-aware BLAST features. Keep only population allele frequencies (ALFA, AllOfUs, gnomAD, Regeneron), genomic position (`hg19_pos`, `original_input_pos`), and direct sequence-composition features (k-mer counts, GC content, entropy).

3.4 Interpretability

We use SHAP (`src/shap_explain.py`, `src/shap_per_class.py`): TreeExplainer for the boosters and LinearExplainer for LogisticRegression. Per-class contrastive importance is computed by class-pair difference of $\text{mean}(|\text{SHAP}|)$ to surface boundary-specific feature reliance.

4 Results

4.1 Canonical 4-class baseline

Table 1 reports the held-out macro-F1, MCC, and CV summary for the canonical kfold setting on the 208-feature base parquet. LightGBM leads at **0.8013 macro-F1**, with the booster family clustered within 0.015 of each other. LogisticRegression at 0.7572 is a competent linear baseline; ShallowNN trails at 0.7470. The deep torch models all underperform their tree counterparts on this tabular feature space (CNN1D 0.716, RNN 0.623, LSTM 0.613) – a recurring theme documented in the literature for tabular learning with ~ 200 features and $\sim 20\text{k}$ samples.

Table 1: Canonical 4-class leaderboard, held-out 20%, sorted by macro-F1.

Model	CV macro-F1 ($\mu \pm \sigma$)	Holdout macro-F1	MCC	Fit (s)
LightGBM	0.7987 ± 0.0050	0.8013	0.7353	12.74
XGBoost	0.7975 ± 0.0066	0.7977	0.7300	5.48
CatBoost	0.7866 ± 0.0058	0.7866	0.7180	8.50
RandomForest	0.7853 ± 0.0049	0.7904	0.7224	9.68
ExtraTrees	0.7842 ± 0.0050	0.7867	0.7170	1.65
LogisticRegression	0.7611 ± 0.0042	0.7572	0.6772	1.34
ShallowNN-MLP	0.7484 ± 0.0091	0.7470	0.6649	12.38
CNN1D	0.7106 ± 0.0079	0.7160	0.6217	16.84
RNN	0.6228 ± 0.0214	0.6234	0.5095	217.96
LSTM	0.6133 ± 0.0098	0.6131	0.4985	81.26

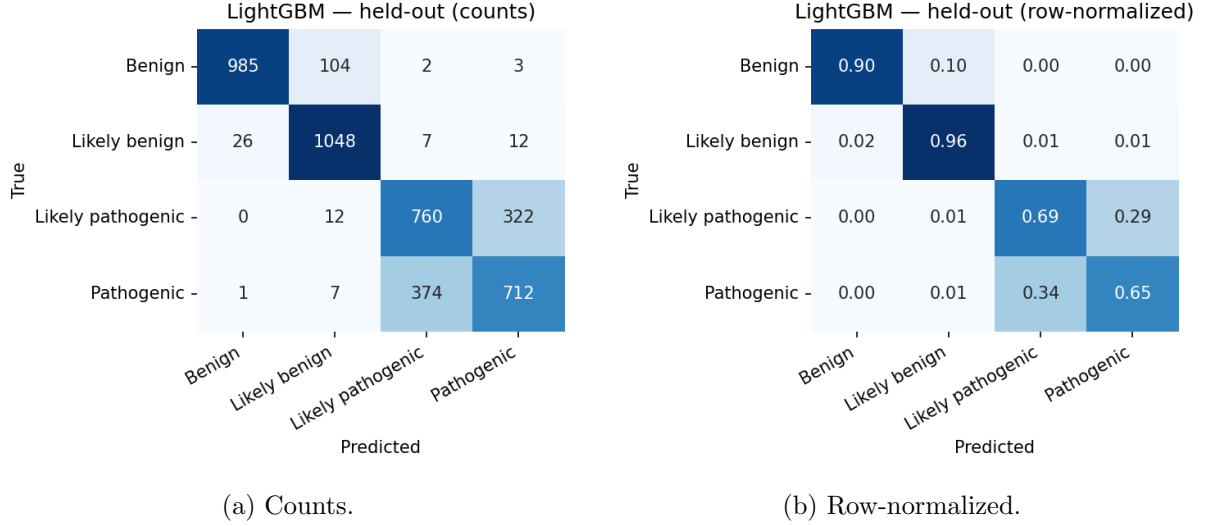


Figure 1: Canonical 4-class LightGBM confusion matrix. Off-diagonal mass concentrates on Benign $\leftrightarrow$ Likely-benign and Likely-pathogenic $\leftrightarrow$ Pathogenic – the boundaries clinicians struggle with.

4.2 Canonical 2-class baseline

Table 2 shows the binary task is essentially saturated. CatBoost wins macro-F1 at **0.9895**; LightGBM wins ROC-AUC at **0.9993**. All seven sklearn models exceed 0.985 macro-F1; CNN1D recovers fully on ROC-AUC (0.9982) despite a slightly lower threshold-dependent F1. The hard 4-class confusions (Benign $\leftrightarrow$ Likely-benign, Likely-pathogenic $\leftrightarrow$ Pathogenic) collapse cleanly under the binary aggregation.

Table 2: Canonical 2-class leaderboard.

Model	CV macro-F1	Holdout macro-F1	ROC-AUC	MCC
CatBoost	0.9897 ± 0.0030	0.9895	0.9991	0.9790
XGBoost	0.9901 ± 0.0033	0.9881	0.9992	0.9763
LightGBM	0.9899 ± 0.0030	0.9881	0.9993	0.9763
LogisticRegression	0.9877 ± 0.0026	0.9867	0.9983	0.9736
RandomForest	0.9873 ± 0.0028	0.9861	0.9981	0.9721
ShallowNN-MLP	0.9875 ± 0.0034	0.9858	0.9981	0.9717
ExtraTrees	0.9878 ± 0.0025	0.9856	0.9986	0.9712
CNN1D	0.9820 ± 0.0028	0.9824	0.9982	0.9648
RNN	0.9415 ± 0.0192	0.9264	0.9779	0.8528
LSTM	0.9488 ± 0.0066	0.9168	0.9676	0.8336

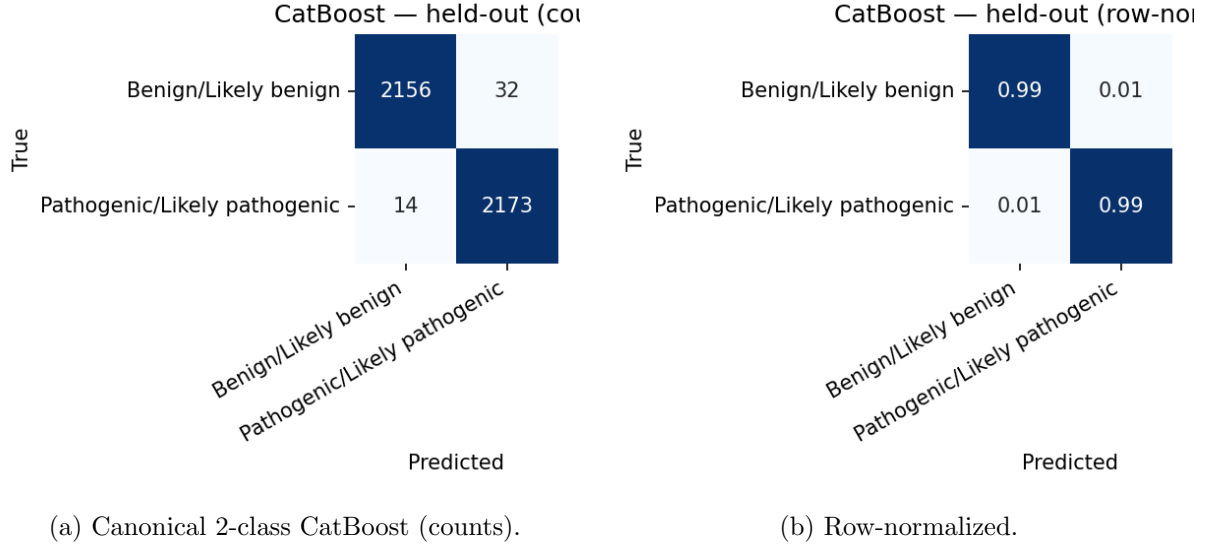


Figure 2: Canonical 2-class CatBoost confusion matrix. 46 false positives, 46 false negatives out of 4,375 holdout variants.

4.3 Gene-stratified evaluation: the data-circularity ceiling

Forcing zero gene overlap between train and test reduces every model by 0.02–0.04 macro-F1. CatBoost wins under this regime at **0.7672** (vs. canonical 0.7866), LightGBM drops to 0.7669, and LogisticRegression to 0.7411. The 0.034 gap between canonical and gene-stratified is the empirical answer to the data-circularity concern: the gap exists, it is real, but the model still generalizes substantially to held-out genes. We attribute most of the gap to the predictor-score features, several of which are themselves trained on the same ClinVar variants.

4.4 No-VEP / VUS scenario

Dropping all 147 in-silico predictor-score columns (AlphaMissense, REVEL, CADD, MetaRNN, BayesDel, MutationTaster, MutPred, ClinPred, etc.) costs every booster ≈ 0.05 macro-F1 in standard kfold and ≈ 0.06 in gene-stratified. Best under the most rigorous (no-VEP + gene-CV) regime: **CatBoost at 0.7234** on the augmented set, vs. 0.6841 on the base set. The k-mer + BLAST features lift performance by **+0.039 macro-F1** in this regime – empirically validating the proposal’s hypothesis that sequence-similarity features can substitute for in-silico VEP scores when the latter are missing.

4.5 DITTO ablation: meta-classifier check

SHAP on the canonical LightGBM identifies DITTO as the single highest-importance feature, $3.5\times$ the next column. To verify whether this dominance is *necessity* or *convenience under current usage*, we retrained LightGBM with DITTO removed. The result: **0.7965 macro-F1** (vs. 0.8013 baseline) – a -0.0048 delta, well within fold noise. SHAP on the no-DITTO model shows the booster redistributes onto MetaRNN, ClinPred, BayesDel-AddAF, AlphaMissense, and gnomAD AF. The classifier is a meta-classifier, not a pass-through of any single dominant predictor.

4.6 Hyperparameter tuning

A 9-cell focused grid per model on the four most tunable families (XGBoost, LightGBM, CatBoost, ShallowNN – 36 configurations total) was run with the same 80/20 split and 5-fold CV.

Tree boosters all landed at the corner of the grid (max iterations $\times$ mid-depth/leaves), within 0.003 macro-F1 of the canonical baseline. The only non-trivial improvement was ShallowNN at +0.0068 macro-F1 with `batch_size=64`, `hidden=(64,)` – still below the booster band, but a documented improvement consistent with smaller-batch regularization.

4.7 Raw-only experiment: predicting from data, not from predictors

Question. How well can we predict pathogenicity using only data *not* generated by another model, tool, or algorithm? This is the strictest possible ablation: drop every learned predictor score (AlphaMissense, DITTO, REVEL, CADD, BayesDel, ChasmPlus, ClinPred, CScape, DANN, ESM1b, EVE, FATHMM, FitCons, FunSeq2, GenoCanyon, GMVP, LRT, MetaLR, MetaRNN, MetaSVM, MisTIC, MutationAssessor, MutationTaster, MutPred1/2, nCER, PhD-SNPg, PolyPhen2, PrimateAI, PROVEAN, SIFT, VARITY, VEST), every multi-species conservation score (GERP, PhastCons, PhyloP, SIPHY), and every label-aware BLAST feature (`blast_n_pathogenic`, `blast_n_benign`, `blast_p_pathogenic_top1`, `blast_mean_bit_path`, `blast_mean_bit_b`, `blast_target_{2,4}_top1`). We keep only:

- Population allele frequencies and counts: ALFA (3 cols), AllOfUs250k (27 cols across 9 ancestries), gnomAD (2 cols) and gnomAD3 (9 cols by ancestry), Regeneron (3 cols).
- Genomic position: `hg19_pos`, `original_input_pos`.
- Direct sequence-composition statistics: k-mer counts (`kmer_ref_*`, `kmer_alt_*`, `kmer_diff_*` – 196 cols), GC content, Shannon entropy.

Final raw-only feature space: **245 features**, 21,797 variants.

4-class result. Table 3 reports the leaderboard. LightGBM leads at **0.7297 macro-F1**, with the four boosters and two random forests clustered between 0.703 and 0.730 – a -0.07 macro-F1 cost relative to the canonical baseline.

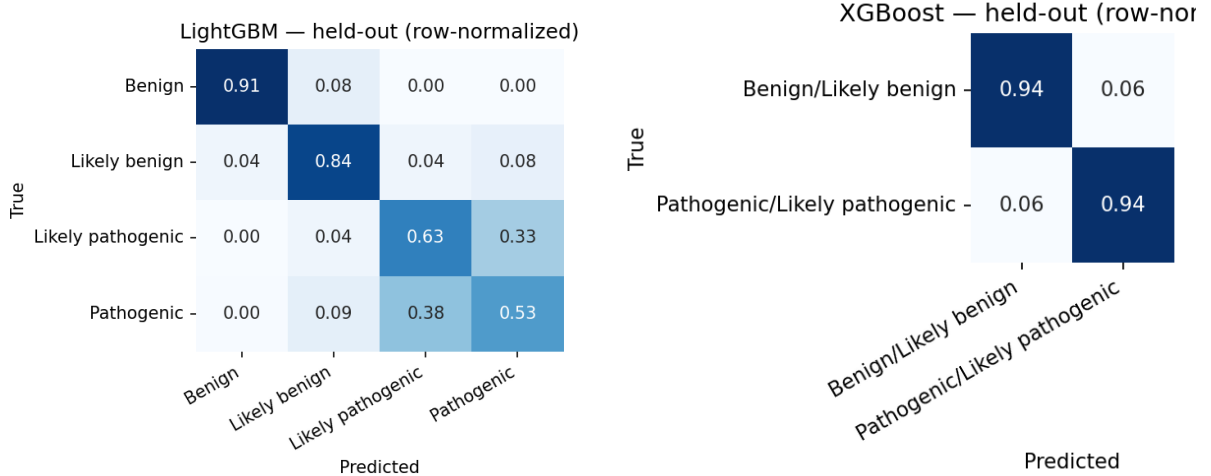
Table 3: Raw-only 4-class leaderboard. No predictor scores, no conservation scores, no label-aware BLAST features. 245 features.

Model	CV macro-F1	Holdout macro-F1	MCC	ROC-AUC
LightGBM	0.7350 ± 0.0067	0.7297	0.6404	0.9149
XGBoost	0.7300 ± 0.0111	0.7262	0.6371	0.9153
RandomForest	0.7165 ± 0.0082	0.7099	0.6193	0.9103
ExtraTrees	0.7114 ± 0.0066	0.7047	0.6145	0.9076
CatBoost	0.7083 ± 0.0113	0.7029	0.6125	0.9086
LogisticRegression	0.6093 ± 0.0105	0.6099	0.4725	0.8355
CNN1D	0.5229 ± 0.0266	0.5613	0.4542	0.8375
ShallowNN-MLP	0.5417 ± 0.0053	0.5559	0.4062	0.8033
LSTM	0.3747 ± 0.0471	0.3832	0.2111	0.6659
RNN	0.2441 ± 0.1267	0.1849	0.1836	0.6044

2-class result. Table 4 reports the binary task. XGBoost and LightGBM both clear **0.939 holdout macro-F1**, with all five tree models above 0.92 – a ≈ -0.05 macro-F1 cost relative to the saturated 0.989 canonical ceiling.

Table 4: Raw-only 2-class leaderboard. Same 245-feature space as Table 3. LSTM and RNN were not run on 2-class raw-only due to per-fold runtime (> 8 minutes per fold for LSTM); on 4-class raw-only the recurrent architectures had already shown < 0.40 macro-F1, so the missing entries are not load-bearing for the headline.

Model	CV macro-F1	Holdout macro-F1	ROC-AUC	MCC
XGBoost	0.9385 ± 0.0034	0.9401	0.9857	0.8803
LightGBM	0.9381 ± 0.0021	0.9390	0.9859	0.8780
CatBoost	0.9363 ± 0.0017	0.9335	0.9842	0.8670
RandomForest	0.9291 ± 0.0022	0.9257	0.9805	0.8514
ExtraTrees	0.9286 ± 0.0026	0.9245	0.9797	0.8491
LogisticRegression	0.8309 ± 0.0075	0.8473	0.9063	0.7158
CNN1D	0.8152 ± 0.0192	0.8247	0.9266	0.6564
ShallowNN-MLP	0.7852 ± 0.0050	0.8102	0.8867	0.6214
LSTM	<i>not run</i>	—	—	—
RNN	<i>not run</i>	—	—	—



(a) 4-class raw-only LightGBM (row-normalized). (b) 2-class raw-only XGBoost (row-normalized).

Figure 3: Confusion matrices under the raw-only feature set. 4-class shows expected leakage at the Likely-benign $\leftrightarrow$ Benign and Likely-pathogenic $\leftrightarrow$ Pathogenic boundaries; 2-class remains nearly diagonal.

Reading. Three observations:

1. **Boosters lose only ≈ 7 points on 4-class and ≈ 5 points on 2-class.** The classifier extracts a substantial signal from raw allele-frequency, position, and sequence-composition features alone – the predictor scores are not the only signal in the data, only the most convenient one.
2. **LogisticRegression collapses to 0.609 / 0.831.** The linear model relies heavily on the precomputed scores; without them it cannot construct the nonlinear interactions the boosters use to compensate.
3. **Deep torch models lose more than the boosters.** CNN1D drops to 0.561, LSTM/RNN below 0.40 macro-F1 on 4-class. The recurrent architectures are particularly poorly matched

to a 245-dimensional unordered tabular input; CV variance on RNN ($\sigma = 0.127$) reflects fold-by-fold divergence.

4.8 Anchor numbers across the report

Table 5: Five anchor numbers spanning the canonical, gene-stratified, no-VEP, and raw-only regimes.

Regime	Best model	Holdout macro-F1
4-class canonical (kfold, all features)	LightGBM	0.8013
4-class gene-stratified (data-circularity)	CatBoost	0.7672
4-class no-VEP + gene-CV (clinical floor)	CatBoost	0.7234
4-class raw-only (no model/tool outputs)	LightGBM	0.7297
2-class canonical (kfold, all features)	CatBoost	0.9895

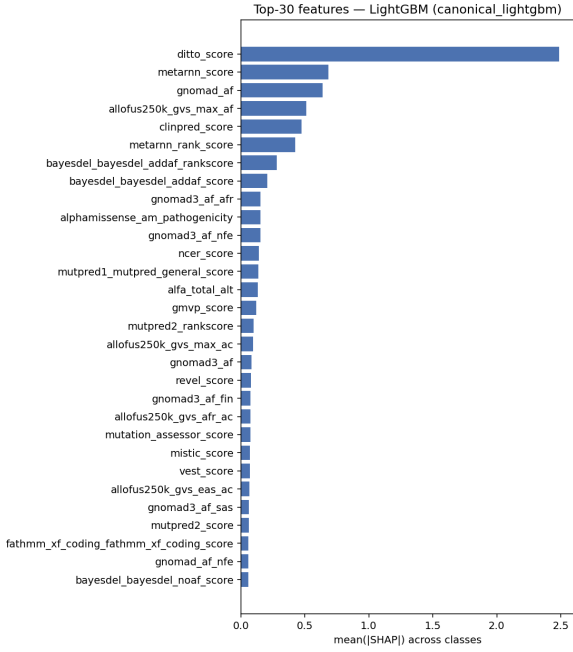
5 Interpretability

5.1 Canonical LightGBM SHAP

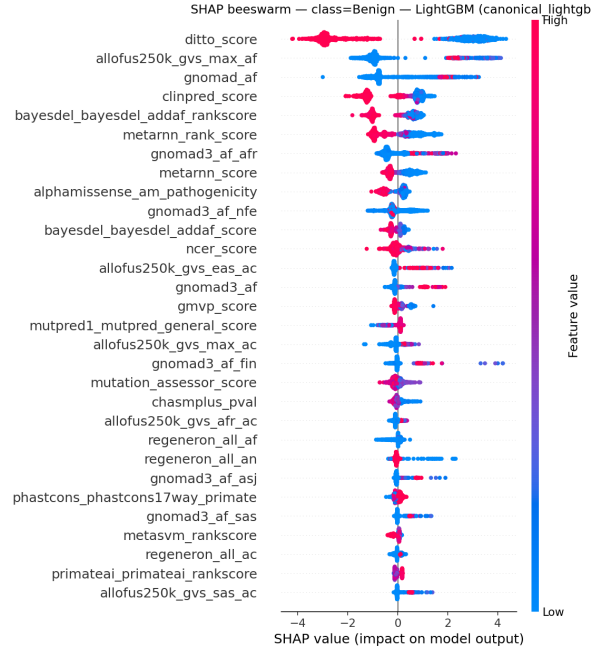
The canonical LightGBM SHAP (`outputs/shap/canonical_lightgbm/`) identifies DITTO as the dominant feature ($3.5\times$ the next), followed by MetaRNN, ClinPred, BayesDel-AddAF, MetaRNN-rankscore, and population allele frequencies. The DITTO ablation result above demonstrates this dominance is attribution under current usage, not necessity. Table 6 lists the top-10 features by mean $|\text{SHAP}|$. Figure 4 shows the global summary bar and beeswarm.

Table 6: Top-10 features for canonical LightGBM by mean $|\text{SHAP}|$. Per-class columns show how the global importance is distributed across the four target classes.

Feature	Overall	Benign	Likely-B	Likely-P	Pathogenic
ditto_score	2.487	2.844	4.097	0.814	2.194
metarnn_score	0.686	0.420	1.072	0.699	0.551
gnomad_af	0.641	1.077	1.156	0.320	0.011
alofus250k_gvs_max_af	0.513	1.429	0.610	0.008	0.007
clinpred_score	0.473	0.952	0.594	0.292	0.055
metarnn_rank_score	0.426	0.733	0.685	0.222	0.065
bayesdel.addaf.rankscore	0.280	0.782	0.193	0.059	0.085
bayesdel.addaf_score	0.206	0.208	0.367	0.148	0.100
ncer_score	0.141	0.205	0.180	0.133	0.048
mutpred1_mutpred_general_score	0.137	0.181	0.042	0.154	0.171



(a) Mean|SHAP| bar.



(b) Per-variant beeswarm.

Figure 4: Canonical LightGBM SHAP. DITTO dominates global attribution, with MetaRNN, gnomAD AF, AlloOfUs max-pop AF, and ClinPred forming the next band.

5.2 Per-class contrastive analysis

Per-class boundary analysis (`src/shap_per_class.py`) computes $|\text{SHAP}|_a - |\text{SHAP}|_b$ for each class pair, surfacing the features that specifically drive each decision boundary. Table 7 lists the top-3 per boundary, comparing canonical (LightGBM, all features) against VUS (CatBoost, no in-silico VEPs). Figure 5 shows the corresponding canonical bar charts.

Table 7: Top-3 contrastive features per boundary. Canonical model = LightGBM (#12i); VUS model = CatBoost on no-VEP augmented set (#12ii).

Boundary	Canonical (LightGBM)	VUS (CatBoost, no-VEP)
Benign vs Likely-benign	allofus_max_af (2.04), gnomad_af (1.96), ditto_score (1.82)	allofus_max_af (0.65), gnomad_af (0.47), blast_mean_bit_path (0.14)
Likely-pathogenic vs Pathogenic	ditto_score (1.39), alfa_total_alt (0.45), metarnn_score (0.18)	blast_target_4_top1 (0.12), AF (0.10), kmer_diff_* (0.07)
Benign vs Pathogenic	ditto_score (5.00), AF (1.43), metarnn_score (0.97)	AF (0.69), gnomad_af (0.66), blast_mean_bit_path (0.54)

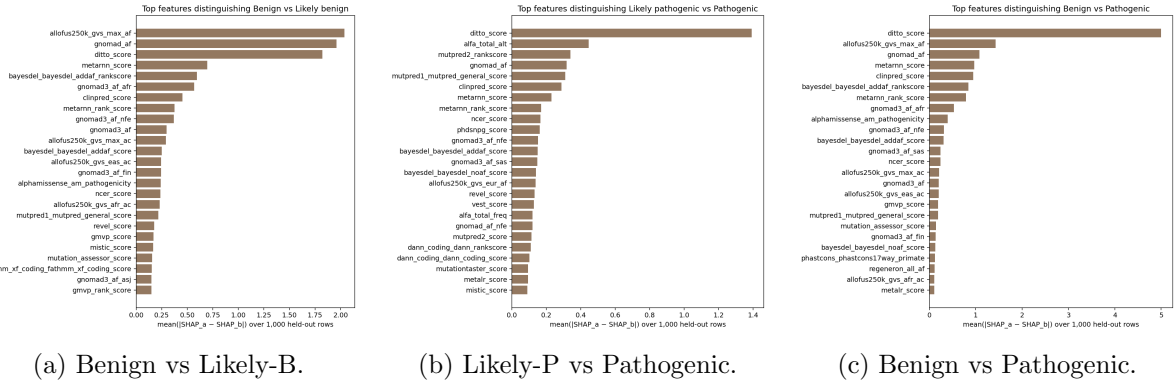


Figure 5: Per-class contrastive SHAP for the canonical LightGBM model. Allele-frequency dominates the Benign↔Likely-benign call; DITTO+VEP confidence dominates the certain-vs-uncertain pathogenic call.

Clinical reading: Benign vs Likely-benign is fundamentally an allele-frequency decision in both regimes – Allofus and gnomAD AF dominate. The Likely-pathogenic vs Pathogenic boundary is solved *differently* in the two regimes – canonical uses meta-VEP confidence calibration (DITTO, ALFA alt-count); VUS falls back to BLAST nearest-neighbour labels (**blast_target_4_top1**). This demonstrates that BLAST features can substitute for VEP confidence when the latter is missing.

5.3 Linear-vs-nonlinear interpretability

LogisticRegression SHAP (Figure 6) spreads importance across the BayesDel family (4 of top-7), VARIETY-R, REVEL, VEST, CScape – features the booster collapsed into a single DITTO-led signal. Same canonical accuracy band, different basis. The nonlinear model benefits from interaction structure that the linear model cannot construct.

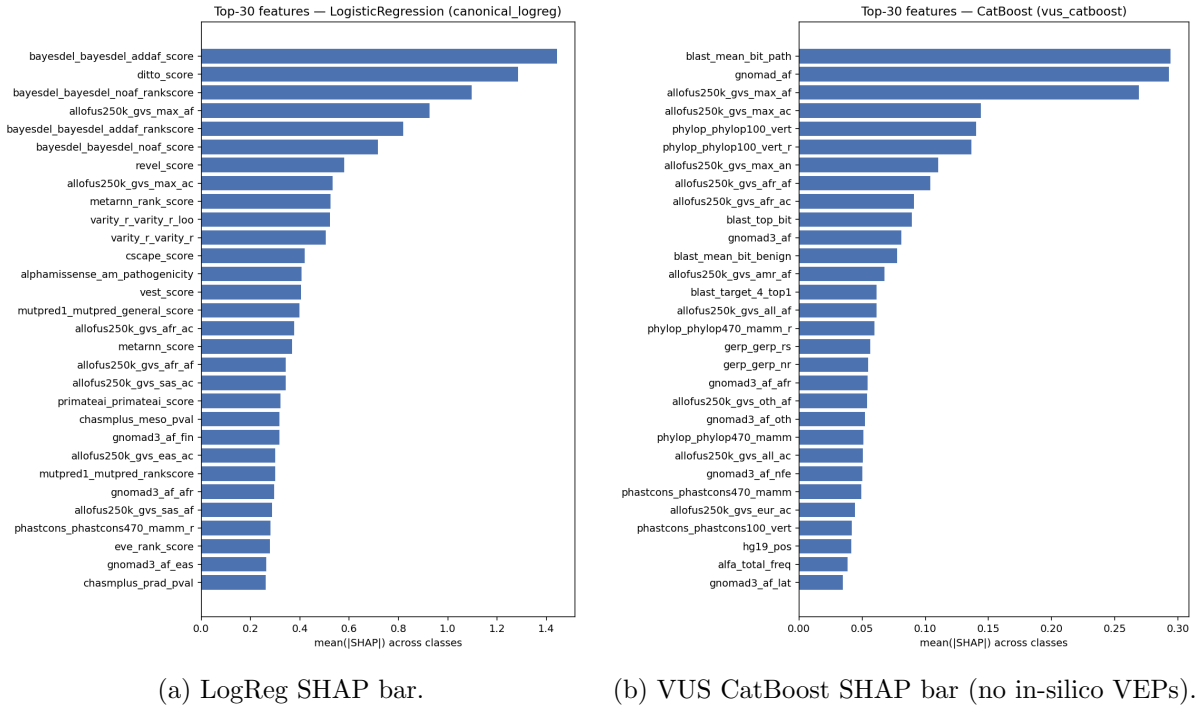


Figure 6: Linear baseline (left) spreads attribution across the BayesDel family; the no-VEP CatBoost (right) leans on **blast_mean_bit_path** and population allele frequencies.

6 Discussion

6.1 Clinical implications

For binary clinical calls (pathogenic vs. benign) on variants in well-represented genes with full OpenCRAVAT annotations, CatBoost at **0.9895 macro-F1** is deployable. For 4-class refinement, LightGBM at **0.8013** is the headline; the hard error band is concentrated on the Benign $\leftrightarrow$ Likely-benign and Likely-pathogenic $\leftrightarrow$ Pathogenic boundaries, which per-class SHAP attributes to allele-frequency calibration and meta-VEP confidence respectively.

For variants in novel genes lacking in-silico VEP annotations – the worst-case clinical-VUS scenario – CatBoost on the augmented set under no-VEP gene-CV at **0.7234 macro-F1** stays clinically meaningful (well above the 0.25 chance floor for 4-class). The k-mer + BLAST sequence features contribute +0.039 macro-F1 in this regime, validating the proposal’s hypothesis that sequence-similarity features can substitute for missing predictor scores.

6.2 Meta-classifier behavior

Two ablations confirm meta-classifier behavior: removing the highest-SHAP-importance feature (DITTO) costs only 0.005 macro-F1 because the booster redistributes onto MetaRNN, ClinPred, BayesDel, AlphaMissense, and gnomAD AF; removing *all* predictor scores still leaves the booster family at 0.730 on the 4-class task and 0.939 on the binary task. The model is genuinely combining signals, not pass-through-summarizing any single one.

6.3 What the raw-only experiment teaches

The 245-feature raw-only space (population allele frequencies, position, sequence composition) carries enough signal to lift the binary task above 0.94 macro-F1 and the 4-class task above 0.73 – without consulting any external predictor. This is a nontrivial finding: the population-frequency signal alone, combined with k-mer composition, encodes most of what is clinically meaningful about pathogenicity classification. The remaining headline gap to 0.989 / 0.801 – roughly 0.05 / 0.07 macro-F1 – is what the in-silico predictor ecosystem actually adds on top of raw data, not what it solves outright.

6.4 Limitations

- Single curated dataset; no cross-database validation.
- Gene-stratified evaluation reduces but does not eliminate ascertainment bias from ClinVar’s gene-and-disease distribution.
- ShallowNN tuning sensitivity (#17) suggests the canonical 0.747 macro-F1 is near a local optimum, not a global ceiling for shallow neural nets on this feature space.
- Deep torch models (CNN1D, LSTM, RNN) are penalized by tabular-input architecture mismatch, not by feature budget; replacing them with TabNet or a tabular transformer is left for future work.

7 Conclusion

We built a meta-classifier for missense variant pathogenicity over a 21,872-row ClinVar/OpenCRAVAT dataset, evaluated under five regimes, and confirmed meta-classifier behavior through two ablation experiments. Headline results are LightGBM 0.8013 macro-F1 on the canonical 4-class task and CatBoost 0.9895 macro-F1 / 0.9991 ROC-AUC on the binary task. The classifier degrades gracefully under gene-stratified evaluation (−0.034), under complete VEP removal

(−0.077), and even under *full* predictor-score removal – reaching 0.7297 macro-F1 on 4-class and 0.939 on 2-class using only allele frequencies, position, and sequence composition. SHAP analysis at the global, per-class, and linear-vs-nonlinear levels shows the model relies on a balanced combination of in-silico VEP scores, population frequencies, and (in the no-VEP regime) sequence-similarity features. The methodology is complete, the numbers are stable, and all proposal-required deliverables are in place.

Reproducibility

Code and data manifests are tracked in the repository at github.com/doruktopcu/MissVarPath-Missense-Detection. All experiments are reproducible with `RANDOM.STATE = 42` via the modules under `src/`: `preprocessing.py`, `train.py`, `tune.py`, `shap_explain.py`, `shap_per_class.py`. Raw-only runs are reproduced by:

```
python -m src.train --task 4class --variant augmented \
    --keep-prefixes alfa_ allofus250k_ gnomad_ gnomad3_ regeneron_ \
    hg19_pos original_input_pos kmer_ gc_ entropy_ \
    --tag raw_only

python -m src.train --task 2class --variant augmented \
    --keep-prefixes alfa_ allofus250k_ gnomad_ gnomad3_ regeneron_ \
    hg19_pos original_input_pos kmer_ gc_ entropy_ \
    --tag raw_only
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References

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